GROUP A · MILESTONE 1

Washbot

Shared laundry, sorted.

washbot-ten.vercel.app



QUICK QUESTION · RAISE YOUR HAND

🙋 Who here is **sharing a washer** with other people?



1 PROBLEM & SCOPE

We asked 18 tenants. Last week...



found the washer **busy** when they needed it



17 / 18



found **someone else's laundry** still inside



17 / 18



left their **own** laundry for 30+ minutes



16 / 18

Everyone knows this. Nobody has fixed it.

Each figure = one tenant who shares a washer · survey, n = 18

1 THE PROBLEM

Laundry is a basic need.
Shared machines get overloaded.



People are busy



Wet clothes can't wait



Housemates don't communicate

And nothing on the market fixes it for shared housing.

1 PROBLEM & SCOPE

How might we help tenants who share one washer or dryer know when it's free, and make sure owners know the moment their load is done, so nobody has to chase anyone?

🎯 Room renters in shared HDB and condo flats, Singapore

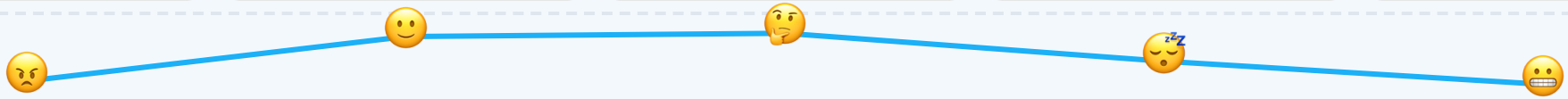
One household · one washer

2 USER & STAKEHOLDER RESEARCH

Two moments break a shared laundry day.

Persona: room renter who shares a washer | **Scenario: an evening wash, start to handover**

Expectation (HMW): know when it's free, owner told when done, nobody chasing

	1 · Check if free ~5 min	2 · Load and start ~5 min	3 · Wait ~1 h cycle	4 · Cycle ends 0–30+ min unnoticed	5 · Next person arrives minutes to hours
DOING	Walks over to look, or asks the group chat	Loads clothes, presses start, leaves	Back in room, guesses the end time	Busy elsewhere, misses the end	Finds clothes inside, asks around, waits or moves them
CHANNEL	At the machine · house group chat	Washer panel	Washer countdown (on site only) · phone timer	None: nobody is told	At the machine · house group chat
THINKING	*"Is it free, or another wasted trip?"*	*"Done, back to my room."*	*"When will it finish?"*	*(not thinking about laundry)*	*"Whose clothes are these? Do I touch them?"*
FEELING					
PAIN POINTS	**Wasted trip when it's busy** 17 of 18 found it busy · 3.3 / 5 Finding 1	**Nothing shows whose load** "Don't know whose load" 3.2 / 5	**No signal, only guessing** 7 of 18 set a phone timer	**Owner is never told it's done** 16 of 18 left their own 30+ min Finding 2	**Free machine, still blocked** 17 of 18 found others' laundry · 10 of 18 ask a chat Finding 2
WORKS WELL	If free, start at once	Quick, one press	Free time while it runs	—	A group chat already exists
OPPORTUNITY	See if it's free without walking over	Show who is using the machine	Know when it will finish	Tell the owner the moment it ends	Remind the owner, so nobody chases

Survey, n = 18 · Thinking is paraphrased from survey patterns, not quotes · times approximate · red = moment of truth

2 USER & STAKEHOLDER RESEARCH

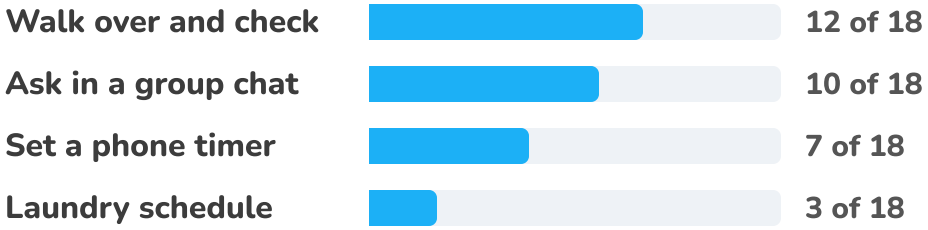
Who we engaged, and who is next.

WHO	HOW	STATUS
Tenants	Survey: 19 responses, **18 answered in full**	Done · 1 Oct
Tenants	Interviews, for quotes and depth	Done
Landlords / master tenants	Interviews: price, and who pays	Next

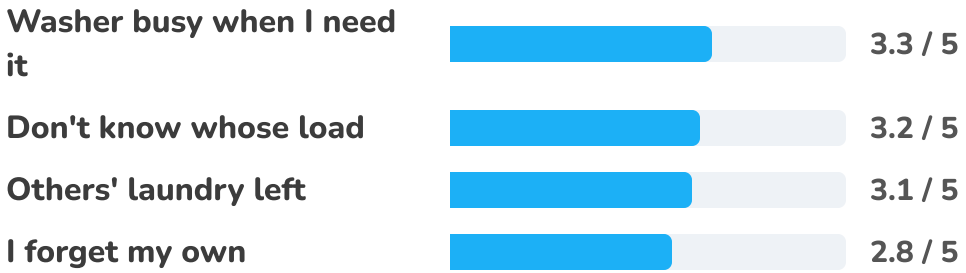
2 USER & STAKEHOLDER RESEARCH

Today's system: walk over and ask around.

How they cope today



How annoying is it?



Survey, n = 18

2 USER & STAKEHOLDER RESEARCH

Three findings.

1 · “Is it free?”

The machine can't tell anyone its state, so 12 of 18 walk over to check.

2 · Victim and culprit

17 of 18 found others' laundry. 16 of 18 left their own. Nobody is told when a cycle ends.

3 · They cope, they'll pay a little

13 of 18 willing to pay S$10 or less.

Survey, n = 18

2 USER & STAKEHOLDER RESEARCH

What the evidence changed.

WE ASSUMED	WE FOUND (N = 18)	SO WE NOW…
Tenants would pay a fair price for a laundry tool	**13 of 18** would pay S$10 or less, and 7 of those would pay S$0. Only 5 would pay more.	Aim for a product **under S$10** per household. Our parts cost has to come down.
Forgotten laundry is the biggest pain	A **busy washer** is rated most annoying: **3.3 / 5**, with 10 of 18 rating it 4–5. Others' left laundry is rated lower (3.1).	Lead with **“is it free?”** so nobody walks over for nothing. Reminders come second.
One or two forgetful housemates cause it	**16 of 18** left their own laundry for 30+ minutes, and 17 of 18 found someone else's.	Design **reminders for everyone**, not blame for one person.
Everyone is on WhatsApp	**13 of 18** use WhatsApp. 5 use another app.	Show status **on the machine itself**, so it works without any app.

3 EXISTING SOLUTIONS & MARKET LANDSCAPE

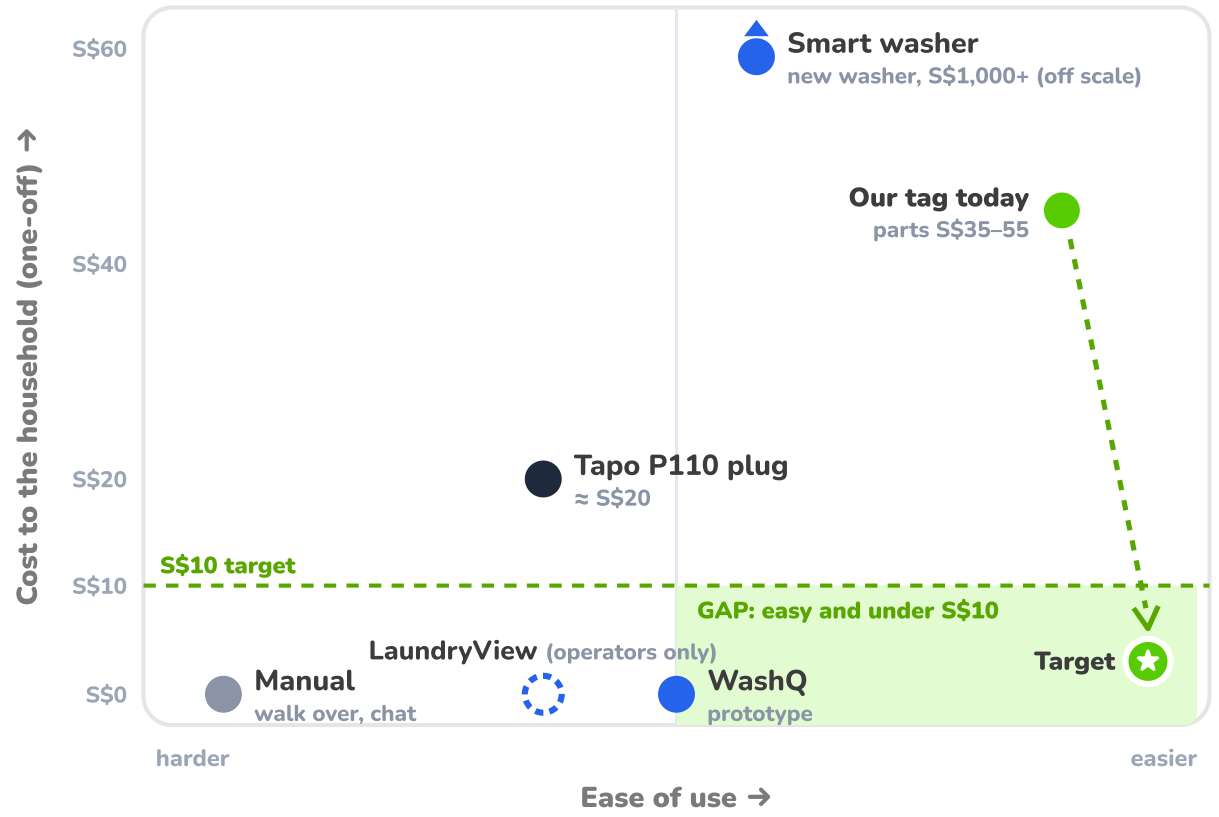
Checking the market options.

OPTION	EASE OF USE	COST	PROBLEMS WITH THIS SOLUTION
Tapo P110 smart plug	**2.5 / 5.** Install an app, pair the plug, plug the washer into it. Only the account owner sees the data.	≈ **S$20** one-off (P110M is S$19.90 on official stores)	Shows power use, not "free or busy". No queue, no names. Above the S$10 target.
LaundryView CSC ServiceWorks	**2.5 / 5.** Only in an operator's laundry room. Find your room on a website, then check it or set an email alert per machine.	**Free** to residents. CSC pays for the install.	Needs the operator's networked machines, so a rented flat can't add it. Shows status, not a queue.
Smart washer SmartThings / ThinQ	**3.5 / 5.** The brand's app shows status and says when the cycle ends. Only people added to the account get it.	**S$1,000+**: buy a new washer	The washer belongs to the landlord, so tenants can't swap it. No queue, no names.
WashQ queue app	**3 / 5.** Web app, no sign-up, but a code for every wash, and every housemate must use it.	**Free** (no price found)	A prototype: only 1 of its 15 demo machines has a live sensor. Play Store listing not verified.
Our product to be tested	**4.5 / 5** (hypothesis). Scan a QR in WhatsApp; the tag shows status on the machine.	≈ S$35–55 parts today. **Target under S$10.**	Too costly today. The hardware has to get much cheaper.

Scores are our assessment from public information, on a 1–5 scale. Not yet rated by tenants.

3 EXISTING SOLUTIONS & MARKET LANDSCAPE

Ease of use against cost.



How we built it

1. **Ease of use:** tenants rely on free workarounds that need no setup (walk over: 12 of 18).
2. **Cost:** 13 of 18 would pay S$10 or less.
3. Cost is the **real one-off price** to a household. Ease is **our 1–5 assessment**.
4. **Pending:** tenants rate each option 1–5.

Read it

● Manual ● Indirect ● Direct

The free options are all a hassle, and the easy one (smart washer) costs S$1,000+. **Our tag is easy but too costly today.** The job is to get under S$10.

Sources: tp-link.com (Tapo, S$19.90) · cscsw.com (LaundryView) · WashQ project page · retail listings (smart washers). Ease is our assessment.

4 NEEDS, REQUIREMENTS & VENTURE HYPOTHESIS

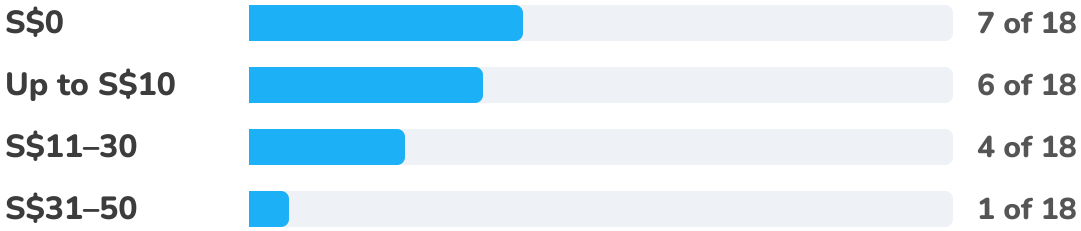
From what they told us to what we must build.

NEED	EVIDENCE (N = 18)	REQUIREMENT	MEASURE
Know if it's free	17 found it busy; 12 walk over	Status visible from anywhere, and on the machine	Wasted trips per week
Laundry gets collected	17 found others' laundry; 16 left their own	Owner told when the cycle ends, then reminded	Minutes from end to collection
Know whose load	8 rate it 4–5 out of 5	Machine shows the current user's name	Loads with a named owner
Nothing new to install	10 already ask a group chat	Works in the chat app they already use	Household using it in week 1
Cheap enough to say yes	13 would pay S$10 or less	One-off cost under S$10 per household	Tenants who accept the price

4 NEEDS, REQUIREMENTS & VENTURE HYPOTHESIS

Tenants will pay a little. Aim: under S$10.

Most a household would pay, once



Survey, n = 18

Early adopter

Tenants who share a washer and already ask around in a group chat.

Value

See if the washer is free and get reminded, for a one-off price under S$10 per household.

Challenge: our rough parts estimate (S$35–55 per tag) is far above S$10. The hardware has to get much simpler, or be shared more widely.

OUR CONCEPT, TO BE TESTED

One idea: Washbot.

A stick-on tag on the washer + a WhatsApp bot. **Scan. Queue. Get pinged.**

▶ Try it live · washbot-ten.vercel.app



APPENDIX · HOW IT WORKS

How the concept works.

1

Scan the tag

Scan the QR on the machine with a phone camera. WhatsApp opens with the message ready to send.

2

Use it or queue

If the washer is free, it is yours. If not, the bot shows the expected wait and books a place in line.

3

Get reminded

A message arrives 5 minutes before your turn and again when the washer is free.

4

Collect and hand over

Reply DONE after taking your clothes out. The next person is told it is their turn.

APPENDIX · THE HARDWARE

A tag that sticks on any machine.



- 📟 20×4 LCD: status, who, when, who's next
- 🟢 RGB light bar: colour at a glance
- 😯 Vibration sensor: knows when it really starts & ends
- 📣 Buzzer: alarm if someone skips the queue
- 🧲 Magnets + USB power: installs in a minute

WHAT WE DO NEXT

Next steps.

1 Opportunity

Add a sourced figure for how many people rent rooms in Singapore.

2 Research

Interview tenants and landlords, then go back to the same people in Round 2.

3 Landscape

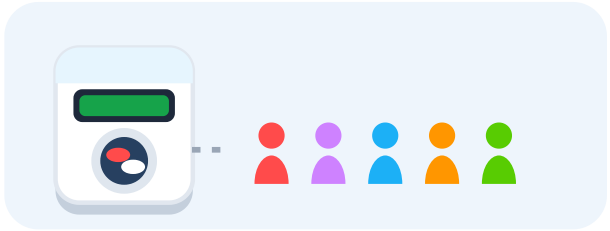
Compare price, effort and time for each alternative with real numbers.

4 Venture

Bring the cost under S$10 and test the price with tenants.

THINKING AHEAD

Where it can grow (hypotheses)



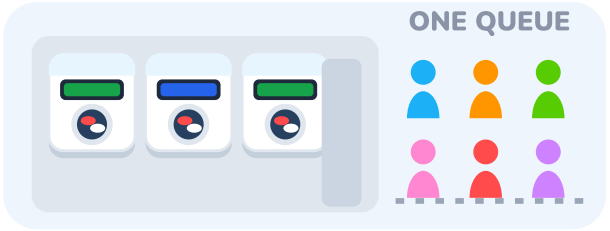
1 Shared rooms

Now. One flat, one shared washer. Where our research is.



2 Many flats

The same tag in many rented flats and condo units. One bot, one WhatsApp number.



3 Dorms and co-living

A laundry room with several machines. One queue for the next free washer.

WASHBOT · GROUP A

Thank you.

Questions? 🙋

Try it: washbot-ten.vercel.app



AFTER THE PRESENTATION

Questions raised in the Q&A.

TOPIC	QUESTION FROM THE AUDIENCE
Privacy and PDPA	What data do we collect, how long do we keep it, and how do we protect it under Singapore's PDPA? How do we pay for storing and protecting it with a S$10 product?
Getting tenants to use it	How do we make sure tenants use the solution, or convince them to?
Tenants who won't use it	What if some housemates don't want to use it, or just don't care? How do we handle non-users and enforce it? Who exactly is the target customer?
Wi-Fi as a penalty	Could it link to the house Wi-Fi and switch it off for people who don't collect their clothes? Suggested for families with children.
Weekday and weekend demand	More people wash at weekends. How is the queue kept fair (first come, first served)? Feedback: suggest the quiet weekday slots.
No internet	If the phone or the machine's tag loses its connection, can people still check the status or book? Should the tag join the house Wi-Fi?
Different cycle lengths	Cycles vary with load size and mode, for example 30 minutes or 60. How do booking and the queue adjust?
Washer and dryer in one round	Today it takes two rounds: one for the washer, one for the dryer. Can the product help do both in one?
Business model	What is the business model, and how do we make a profit from a product that costs only S$10?

Recorded at the Milestone 1 presentation. Open questions for us to work through next.